

REMOTE CONTROL Over Internet

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Switching a device on or off from anywhere in the world is no longer a futuristic concept—it is practical and achievable using a simple Wi-Fi-enabled setup. Whether activating a surveillance camera, an air-conditioner, or a garden sprinkler, remote control opens up endless possibilities limited only by the imagination.

In this system, a computer with internet access communicates with an ESP32 Wi-Fi module connected to the target device. The ESP32, placed within the range of a local Wi-Fi network, serves as the interface. One of its I/O pins is linked to the device—in this case, an LED with a series resistor connected to GPIO4—and can be toggled remotely via a laptop or desktop.

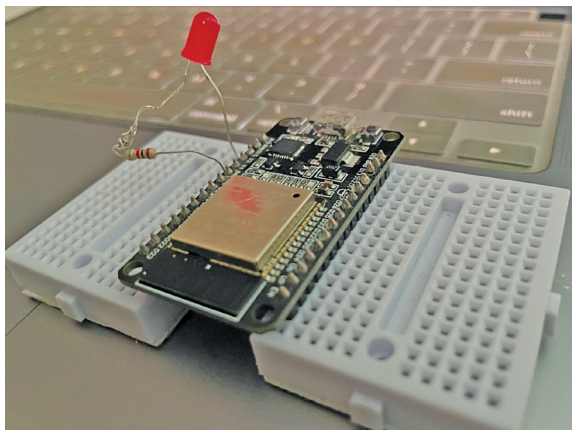


Fig. 1: Author's prototype

The ESP32 operates at 3.3V logic levels, meaning its high output corresponds to 3.3V and all inputs must comply with the same voltage standard. Fig. 1 shows the author's prototype. The required components are listed in the Bill of Materials table.

Circuit diagram

Fig. 2 shows the circuit diagram for the remote control over internet system. It is built around an ESP32 board (MOD1) and a few other components. The ESP32 board features 2.4GHz dual-mode Wi-Fi and Bluetooth, powered by a dual-core processor, and is connected via a USB cable.

The ESP32 can be powered either through the USB cable (typically used while uploading the code) or via the Vin pin by supplying a regulated voltage between 5V and 12V. It is advisable to use a voltage close to 5V to minimise power dissipation in the onboard voltage regulator. Power

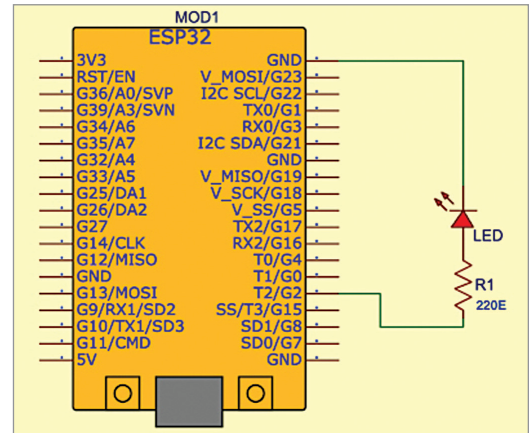


Fig. 2: Circuit diagram

should not be supplied through both the USB and Vin pin simultaneously, as this may damage the board. (Note: A convenient way to power the ESP32 is by using a power bank with a USB port.)

Software

In this setup, two nodes are required—one running on the ESP32 device and the other on the client side, developed in C.

ESP32 Installation. To install the ESP32, first ensure the USB to UART bridge driver is installed, then connect the ESP32 to your computer using a USB cable (without any other power source). Check Device Manager under 'Other devices' for the Bridge Controller and install the necessary driver, noting the assigned COM port. In the Arduino IDE, go to File > Preferences, and under Additional Board Manager URLs, add:

https://espressif.github.io/arduino-esp32/package_esp32_index.json

BILL OF MATERIALS

Item	Description	Quantity
ESP32 board (MOD1)	2.4GHz dual-mode Wi-Fi, Bluetooth dual-core with USB cable	1
LED 5mm	For visual indication	1
220Ω	Current limiting resistance	1